

Part 1 – Project Proposal

Student Name	
Student Number	
Degree Pathway (course)	Bachelor of Science with Honours Computer Science
Supervisor Name	
Title of Project	Smart Study: AI Base collaborative Learning and exam platform
Abstract of the project	The project will involve developing a new mobile app to help students study in groups and remember what they studied. The app will allow students to arrange their work in different subjects and topics, where they can store their notes and create multiple-choice quizzes. The best feature of this app will allow students to share their work. They can choose to share their work privately among friends or publicly. The main reason for developing this project will be that most students will feel lonely while studying, and they forget everything they studied quickly. The mobile app will be made using Flutter for making front ends, so it can run on mobile devices. The back end of the app will be made using Flask Python Framework. The AI tool to read the student's notes and create quizzes from them will be one of the most important unique features. Further, the app will support both self-exams and group exams to test themselves under competitive conditions. The project is unique because it will utilize the capabilities of AI to automate and socialize simultaneously. Most apps will require students to manually create questions, while the proposed app will utilize the capabilities of AI to speed up the process. This study tool will be different from others, as it allows friends to take the test at the same time to compare their performance. There will be a smart notification system that uses an algorithm to remind students to retake quizzes before they are likely to forget what they learned. The project will be important to students because it will save their time, increase motivation, and allow them to see their improvement by utilizing a performance dashboard. It will be able to prove the power of collaboration between AI and social groups to improve educational outcomes.
Project deliverables	Software • Mobile application (Android) • Server-Side flask application Documents • Project proposal • Contextual report • Thesis report • Reflective report
Description of your artefact	Background Most students face difficulties in maintaining their motivation to study alone, and they also face problems in recognizing the most important information from what they have noted down. Traditional methods of study are often passive and lack the required engagement to retain information in long-term memory. Aim To develop a mobile application that can help students stay motivated and retain their lessons for a longer period by using AI to generate learning materials automatically and social aspects that enable friends to learn and compete together.

	Objectives: To develop an interface for the students to learn and manage notes and subjects in an easier way by using the Flutter mobile app. To develop a reliable back end for the app by utilizing the Flask Framework. To Develop an AI tool to read the students' notes and generate accurate questions in multiple-choice format. To develop an exam feature to allow students to compete together in real time. To develop a smart notification system to track students' progress and remind students to learn specific topics. To test the accuracy of the AI-generated questions and the app's overall performance with bug tracking. To collect user feedback from students and find out whether the social and AI features had any actual motivational effects on their learning habits.				
Risk analysis	Risk	Impact	Likelihood	Mitigation Action	
	Students might cheat during the group exams.	Medium	High	Give each question a brief amount of time and mix up the sequence of the questions so that students are unable to copy from each other.	
	The AI makes wrong or confusing quiz questions.	High	Medium	Implement a functionality enabling students to modify or remove a question if the AI system made mistakes	
	Running out of time to finish all the social features.	High	Medium	Implement the Agile approach to complete the most critical functionalities first, such as the login functionality and add notes and quiz.	
	Losing student notes or quiz scores due to a system error.	High	Low	Implement the backup functionality to back up data at regular intervals.	
	The app runs too slowly when many friends join an exam.	Medium	Medium	Optimize the code and test the group exam functionality with a small number of users.	
	The user interface is too difficult for students to use.	Medium	Low	Test the app with a few classmates very early on and modify the design accordingly.	

How does your project relate to your degree course and build upon the units/knowledge you have studied/acquired	Development of this app using Flutter improves the knowledge and skills in mobile development and improves knowledge in backend systems through the implementation of the flask framework. Additionally, it covers Server-Side Load Balancing strategies to effectively handle the number of concurrent users during group examinations and utilize Software Design Patterns to ensure the robustness of the application. The use of Natural Language Processing to analyse the materials for generate quiz improve the level of expertise in AI. Furthermore, this also improves Project Management skills by using of Agile and Scrum Methodologies to ensure better delivery and UI/UX Design using user-centric design to ensure maximum levels of engagement among academics.	
Resources required in developing the artefact	Software - Visual Studio Code, Android Studio, Postman, Git Frameworks – Flutter (Dart), Flask framework Hardware – Laptop, Android base Mobile Phone Data - Academic notes and Study materials, open-source datasets (SQuAD)	
Have you completed & submitted your ethics form?	YES	NO
If the project is a development of previous work by yourself or others, give details below. Failing to declare such previous work here may be treated as an academic offence		

Supervisor Signature: *Nasas*

After the proposal has been signed off by both the supervisor and course coordinator scan the proposal and upload on BREO with signatures. Projects that follow proposals that have not been approved may be cancelled and there will be no compensation for any time lost

Part 2 – List of relevant resources

Fill in this section *after* your project proposal has been approved by your supervisor. Use Harvard referencing (see <https://lrweb.beds.ac.uk/a-guide-to-referencing>). Modify the list below as appropriate. This list is part of Assignment 1 and will be submitted with the Project Proposal.

1. Books

- a. Napoli, M.L., 2019. *Beginning flutter: a hands on guide to app development*. John Wiley & Sons
- b. Grinberg, M., 2018. *Flask web development*. "O'Reilly Media, Inc."

2. Journal Papers

- a. Hamari, J., Koivisto, J. and Sarsa, H., 2014, January. Does gamification work?--a literature review of empirical studies on gamification. In *2014 47th Hawaii international conference on system sciences* (pp. 3025-3034). Ieee.
- b. Pahl, C. and Jamshidi, P., 2016. *Microservices: A Systematic Mapping Study*. CLOSER (1), pp.137-146.

- c. Shukla, A., Kumar, S. and Singh, H., 2018. Load balancing approaches for web servers: A survey of recent trends. *International Journal of Engineering*, 31, pp.242-248.
3. *Web Sites with relevant information*
 - a. *Set up and Test Drive Flutter documentation. Available at:*
https://docs.flutter.dev/install/quick?_gl=1%2A1ywwcpj%2A_ga%2AMTg0NDcyNjk3NS4xNzcxOTUwMjcz%2A_ga_04YGWK0175%2AczE3NzE5NTAyNzQkbzEkZzAkdDE3NzE5NTAyNzQkajYwJGwwJGgw (Accessed: 24 February 2026).
 - b. *Welcome to flask¶ (2026) Welcome to Flask - Flask Documentation (3.1.x).*
Available at: <https://flask.palletsprojects.com/en/stable/> (Accessed: 24 February 2026).
4. *Relevant software*
 - a. *Microsoft (2021) Visual studio code - the open source AI code editor, RSS. Available at: <https://code.visualstudio.com>*
 - b. *POSTMAN (2020) The world's leading API platform: Sign up for free, Postman.*
Available at: <https://www.postman.com>
 - c. *Gunicorn (2020) Serve python on the web, Gunicorn. Available at: <https://gunicorn.org/>.*
 - d. *microsoft (2020) GitHub · change is constant. github keeps you ahead., GitHub.*
Available at: <https://github.com/>.
 - e. *Download Android Studio & App Tools (2015) Android Developers. Available at: <https://developer.android.com/studio>.*
5. *Relevant hardware*
 - a. *Personal computer*
 - b. *Personal mobile phone (Android)*
6. *Other*
 - a. ...
 - b. ...